Configuring Basic Network Hardware Theory

Practical Assignment

Aim

To learn how to access a Wi-Fi router, check network settings, test a static IP address, and locate the DHCP range.

1. Router Admin Login Page

Connect your laptop or phone to your home Wi-Fi. Open a browser and enter the router address, commonly 192.168.0.1 or 192.168.1.1. If it does not open, check the Default Gateway in your device's network settings. On Windows, run **ipconfig** in Command Prompt. When the login page appears, take a screenshot for your practical record. Do not share your router password.

Screenshot: Attach your own router login-page screenshot here.

2. Find Your Device's Network Settings

While connected to Wi-Fi, record your IPv4 address, subnet mask, and default gateway. On Windows, run **ipconfig** and look under the Wi-Fi adapter. On a phone, open the connected Wi-Fi network's details.

Setting	Meaning	Your value
IP address	Address assigned to your device on the local network.	
Subnet mask	Identifies the network portion of the IP address.	
Default gateway	Usually the router's local IP address.	

3. Test a Static IP Address

DHCP assigns network settings automatically. A static IP is entered manually. First write down your current settings. For example, if your router is 192.168.1.1 and the subnet mask is 255.255.255.0, 192.168.1.50 may be a suitable example only if it is unused and does not conflict with the DHCP pool. Use your actual network settings.

On Windows, open Settings → Network & Internet → Wi-Fi → connected network properties → IP assignment → Edit. Select Manual, enable IPv4, enter the correct address, subnet mask or prefix, gateway, and DNS settings, then save. Test internet access. When finished, change IP assignment back to **Automatic (DHCP)**. If the connection fails, restore DHCP immediately.

4. Find the Router's DHCP Range

In the router admin panel, look under LAN, Local Network, or DHCP Server. Record the start and end addresses shown by your router. Example only: 192.168.1.100 to 192.168.1.200.

DHCP setting	Value from your router
Start address	
End address	

What happens if two devices have the same IP address?

An IP conflict occurs when two devices use the same IP address on the same network. One or both devices may lose reliable network access. DHCP normally avoids duplicate assignments, but a manually configured static IP can cause a conflict if it is already in use or overlaps with the DHCP range.

Conclusion

This practical covers accessing a router login page, recording IP settings, testing a static IP, and finding the DHCP range. Always restore Automatic (DHCP) after testing and only change settings on a network you are authorised to manage.